

25th European Conference on Computational Biology (ECCB)

Tutorial session T13: 13:45 -17:30

**Democratizing multi-omics data analysis
with BiomiX and Nextflow: from single-
omics to advanced integration**

Presenters:

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BIOMIX

MULTIOMICS DATA INTEGRATION



Part 2 – Multi-Omics Integration

15:30 – 16:00 | Integration strategies – Jessica Gliozzo

- Early, intermediate and late integration
- Overview of MOFA, DIABLO, SNF and NEMO

16:00 – 16:30 – MOFA & DIABLO Tutorials – Cristian Iperi

- Parameter selection
- Application to the CLL dataset
- Interpretation of results

16:30 – 17:00 – SNF & NEMO Tutorials – Jessica Gliozzo

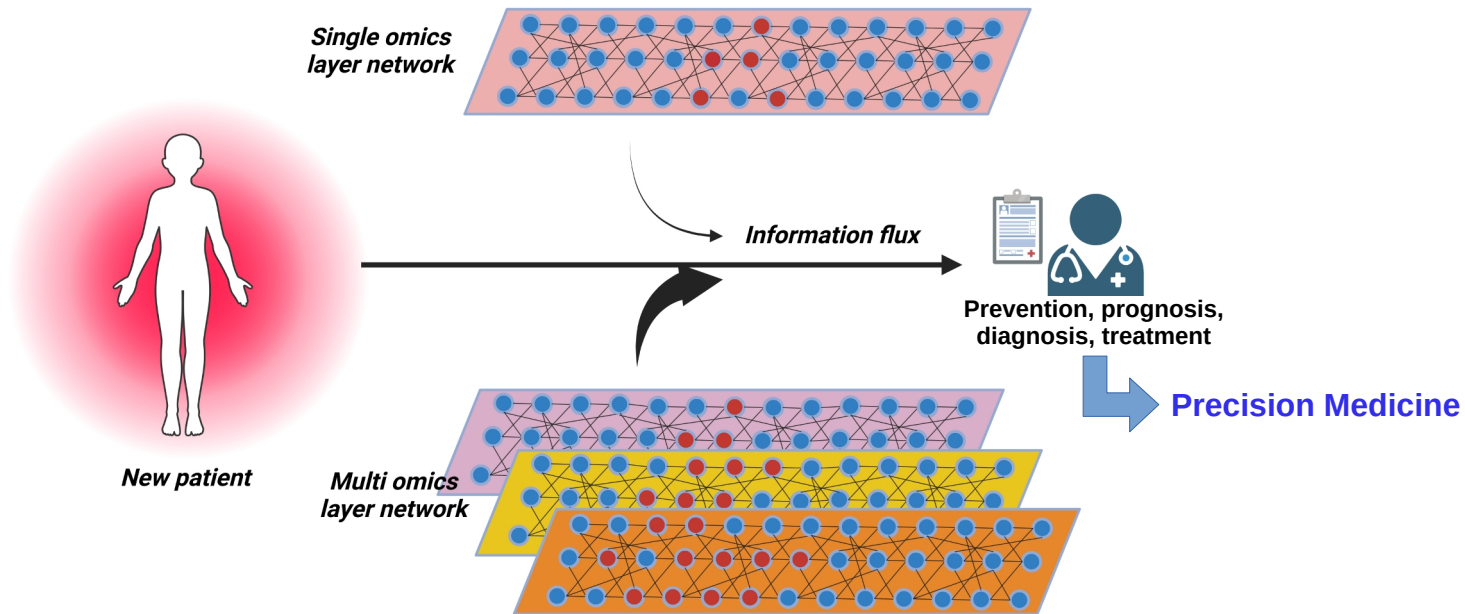
- Parameter selection
- Application to the CLL dataset
- Interpretation of results

17:00 – 17:30 – Final discussion and Q&A





Multi-omics data integration, the grail of biology understanding ?



Pros

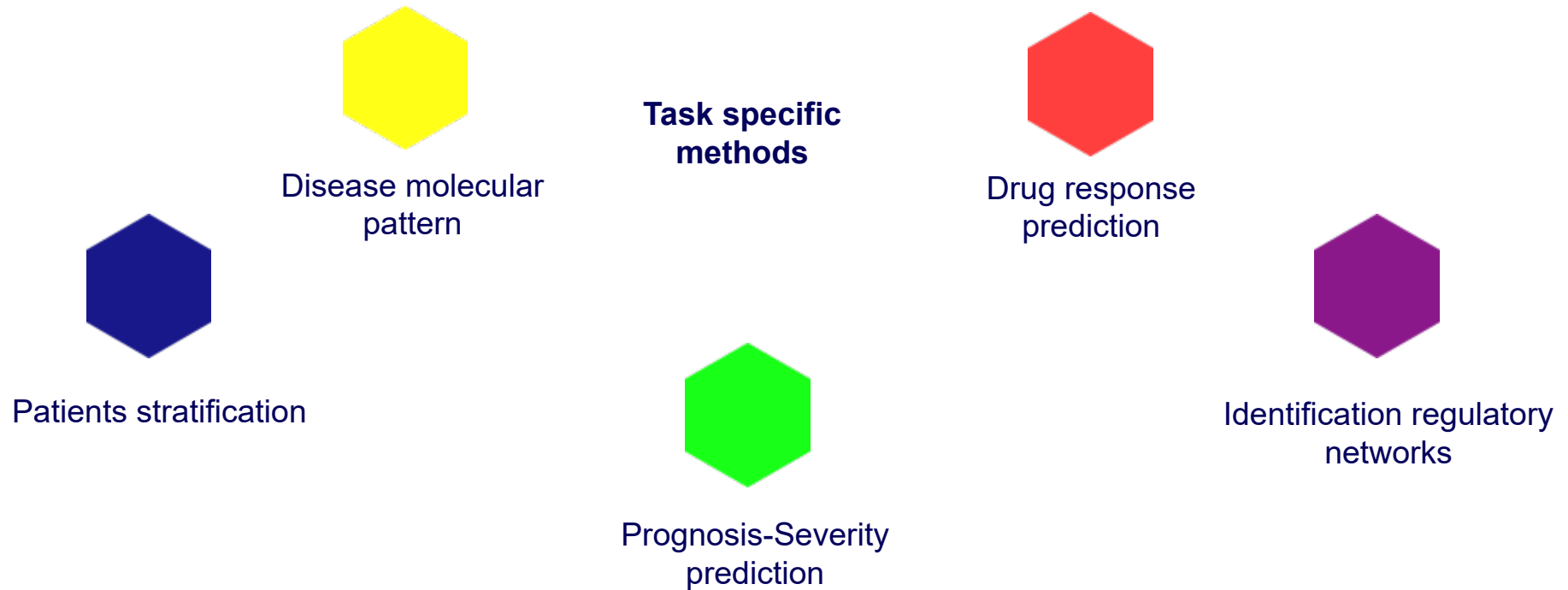
- Observations strengthened
- Holistic interpretation
- Detailed characterization

Cons

- Different distribution
- Curse of dimensionality
- Missing data

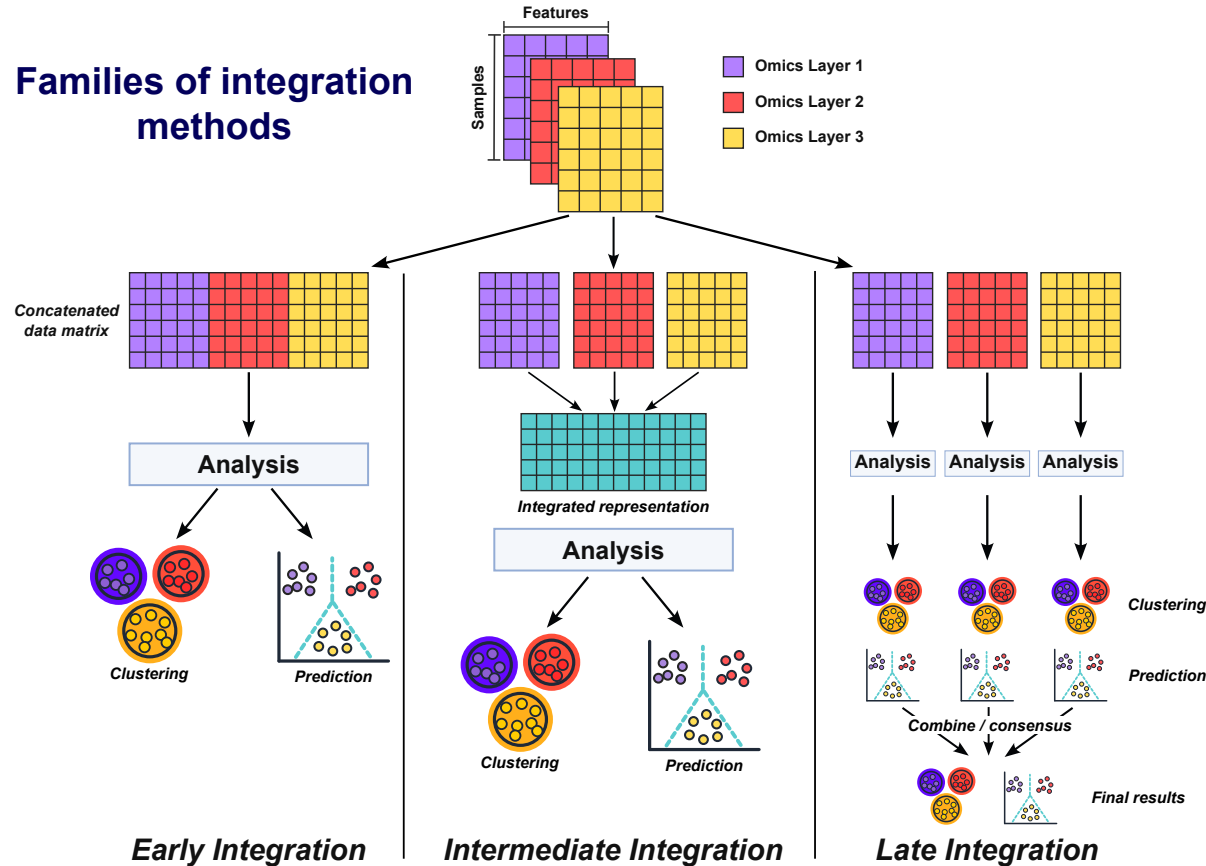


Multi-omics data integration, the grail of biology understanding?





Families of integration methods



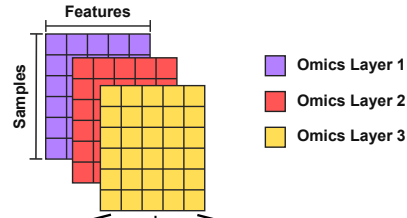
Key distinction is the **stage** at which information from different omics is integrated:

- **Early** - omics are merged into a single input feature space before joint analysis
- **Intermediate** - omic-specific representations are integrated to construct a joint representation
- **Late** - results from separate analyses/models are combined

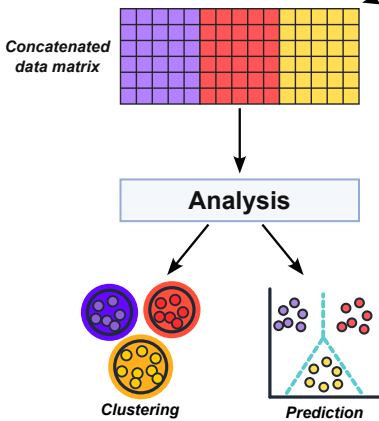
Ritchie MD et al., Nat Rev Genet, 2015; Zitnik M et al., Inf Fusion, 2019



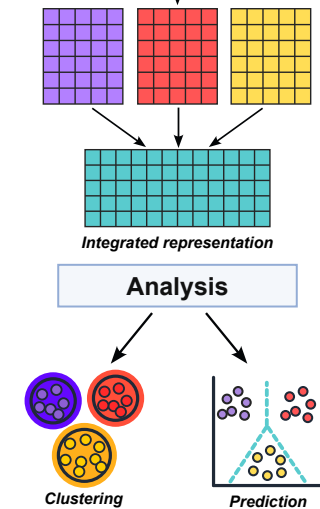
Families of integration methods



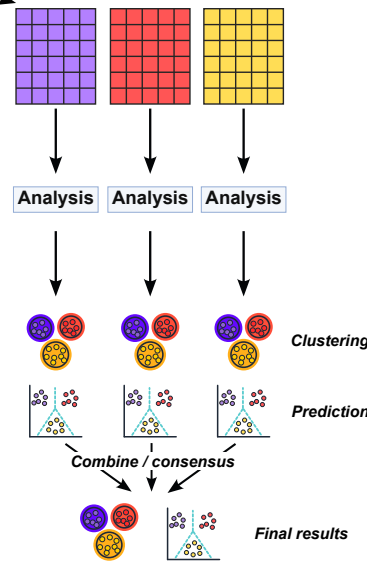
Omics Layer 1
Omics Layer 2
Omics Layer 3



Early Integration



Intermediate Integration



Late Integration

Early integration:

- Different omics measured on the same samples
- Features from the omics are concatenated into a single matrix
- A joint downstream analysis is performed on the integrated matrix

Integration occurs at the input level

Pros

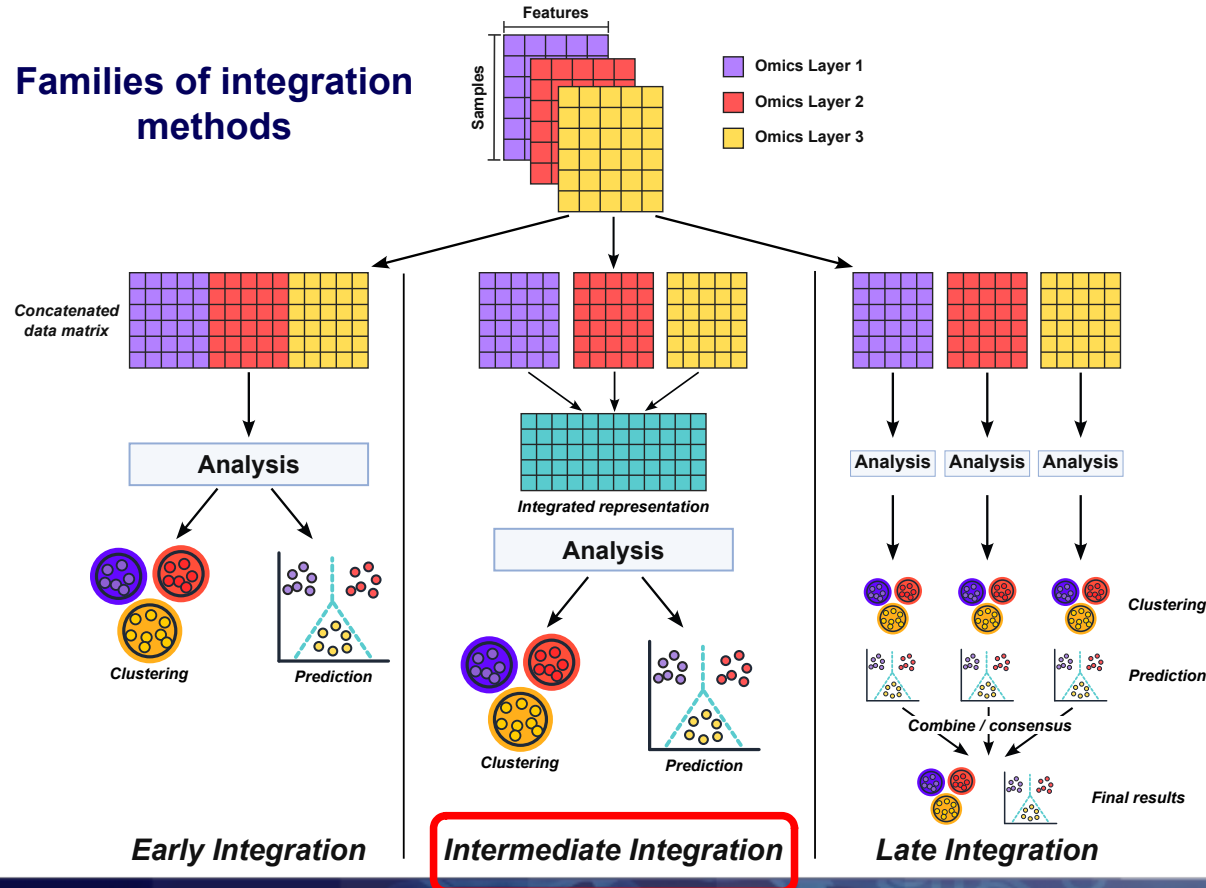
- Simple and intuitive
- Compatible with many standard ML/statistical methods
- Allows joint analysis of features across omics

Cons

- Very high dimensionality
- No explicit modeling of cross-omic structure by concatenation
- Sensitive to differences in scale/distribution across omics
- Often requires matched samples



Families of integration methods



Intermediate integration:

- Omics are retained as separate data layers
- Information across layers is integrated while learning or constructing a joint representation
- The resulting integrated representation is used for downstream analysis

Integration occurs at the representation level

Pros

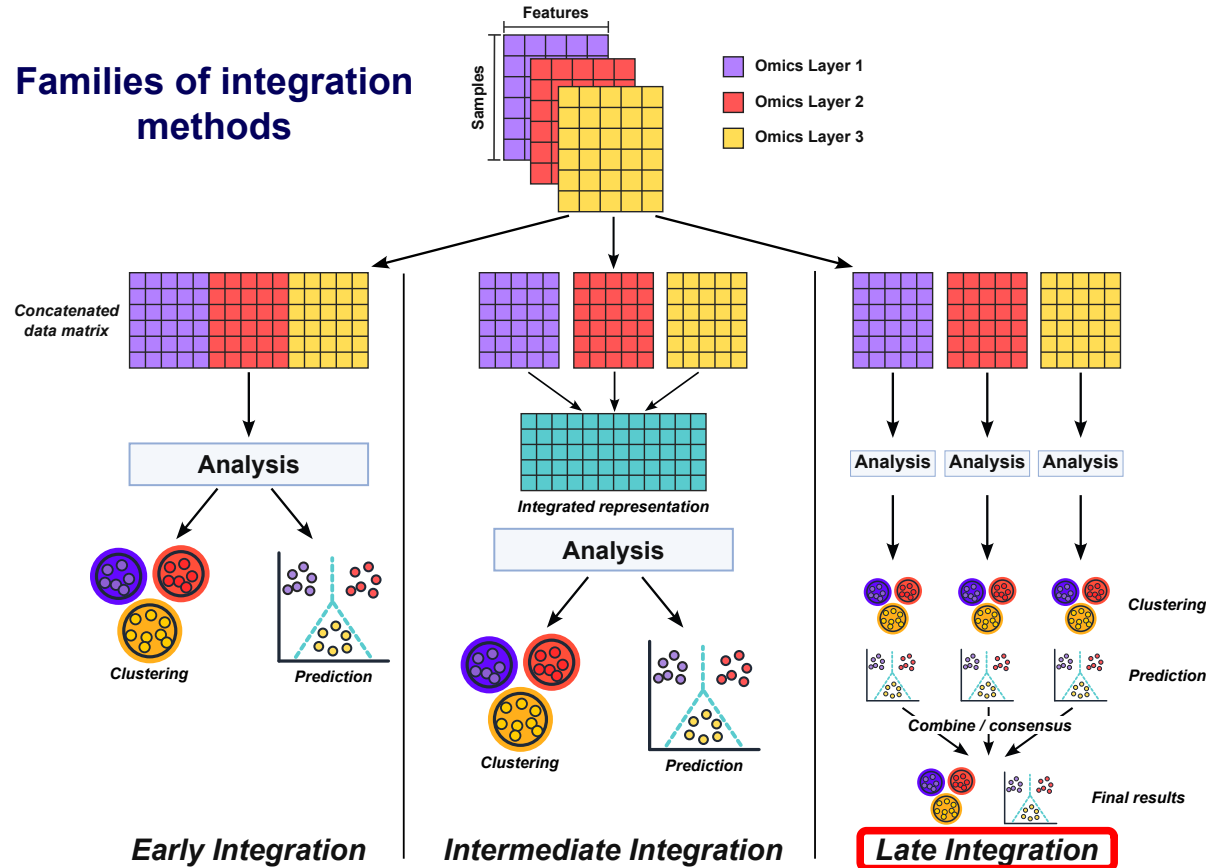
- Can capture relationships across omics
- Can accommodate different data types and distributions
- May reduce the influence of very high-dimensional feature spaces
- Some approaches handle missing samples

Cons

- More complex than simple concatenation
- Results depend on how omic-specific representations are constructed
- Parameter selection can strongly affect the integrated representation
- Interpretation may be less direct



Families of integration methods



Late integration:

- Each omic layer is analyzed separately
- Each analysis produces an omic-specific output (e.g. predictions or clustering solutions)
- The separate outputs are then combined and used to derive a consensus
- The combined output provides the final result

Integration occurs at the output level

Pros

- Allows omic-specific preprocessing and models
- Can accommodate heterogeneous data types and distributions
- Individual omic contributions can remain relatively interpretable
- Some approaches can more easily handle missing modalities
- Separate analyses can be performed independently or in parallel

Cons

- Cross-omic relationships are not modeled directly
- Information may be lost when each omic is reduced to an output
- Results depend on the strategy used to combine or weight outputs
- Signals that are weak within individual omics but strong jointly may be missed



Classification based on computational methods

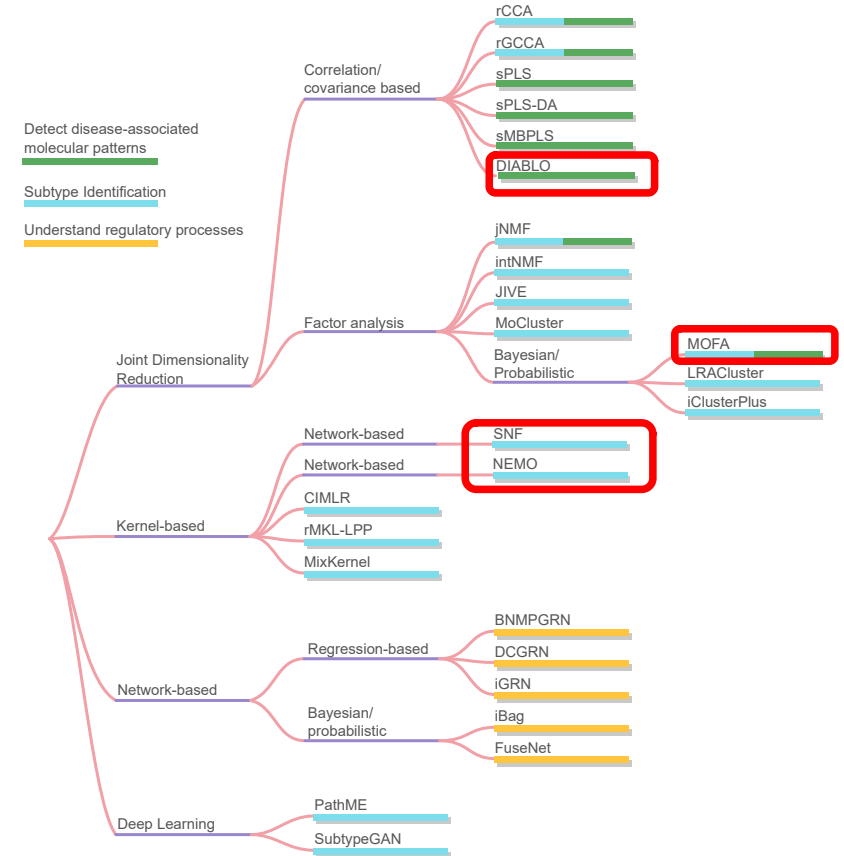
Most newly developed tools specifically adopt the **intermediate integration** approach.

Methods can be classified based on the employed **computational approach**.

In BiomiX, the following intermediate approaches are implemented:

- **MOFA** (Multi-Omics Factor Analysis)
- **DIABLO** (Data Integration Analysis for Biomarker discovery using Latent cOmponents)
- **SNF** (Similarity Network Fusions)
- **NEMO** (NEighborhood based Multi-Omics clustering)

Image adapted from Athieniti E, Spyrou GM, Comput. Struct. Biotechnol. J, 2023

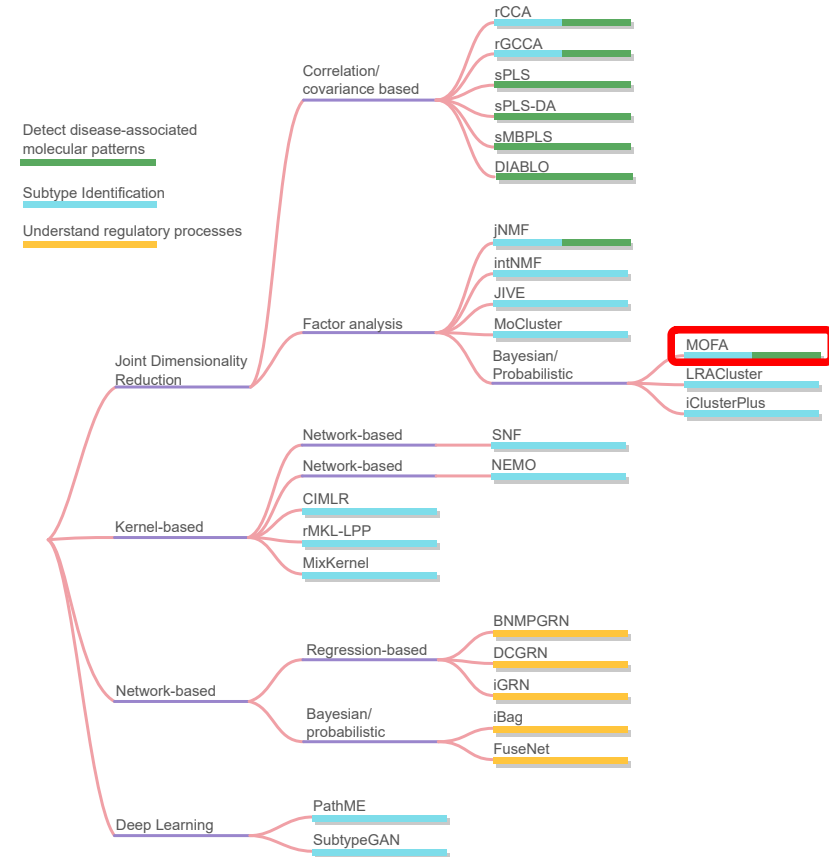




MOFA (Argelaguet R, et al. Mol. Syst. Biol., 2018)

MOFA is an **unsupervised Bayesian factor-analysis framework** that learns a low-dimensional representation of multi-omics data through latent factors **capturing shared and modality-specific sources of variation**.

- Data **integration suited for** Gaussian, Bernoulli and Poisson **distributions**
- **Missing omics** data handled
- **Easy MOFA factor interpretation** enabled by sparse weights
- Low performance on strongly non-linear data
- Factor selection depends on user-defined training criteria





Multiple omics matrices Y^m are approximated using a **common sample-factor matrix Z** , but each omic has its own **feature-weight matrix $W^{(m)}$** .

$$Y^{(m)} \approx ZW^{(m)T} \quad m = 1, \dots, M$$

$Y^{(m)}$: one data matrix for each omic m

Z : one common factor matrix for the samples

$W^{(m)}$: one loading/weight matrix for each omic m

Factor-wise and feature-wise **sparsity** lead to more interpretable solutions!

Each factor can capture variation **shared** across multiple omics or **specific** to a single omic.

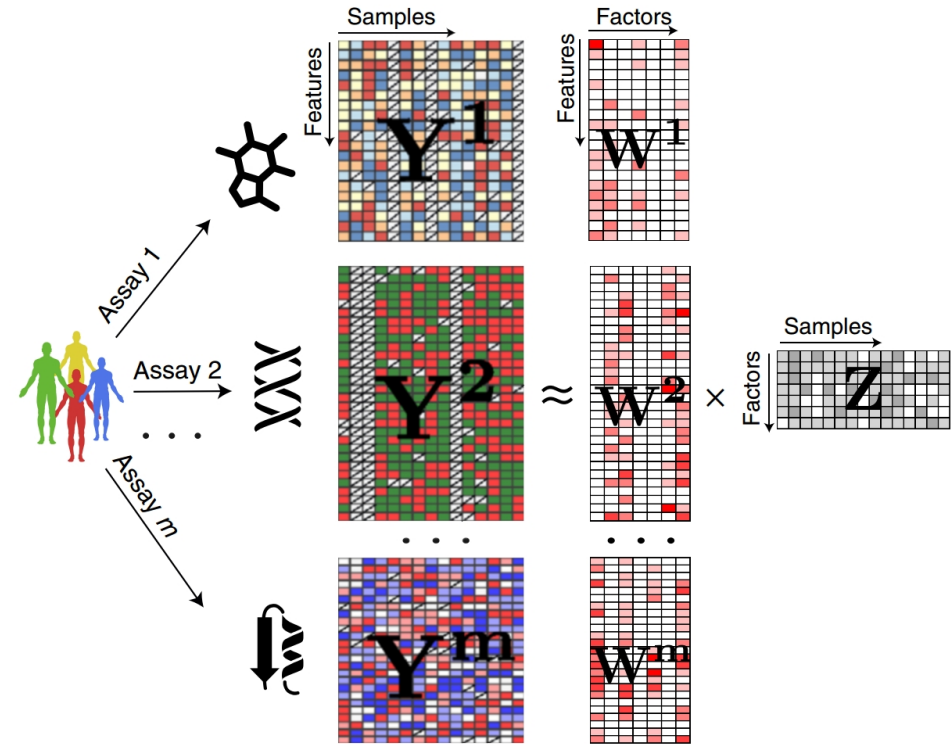


Image from Argelaguet R, et al. Mol. Syst. Biol. 2018



Making it simple: Intuition behind MOFA

MOFA can be viewed as a **multi-omics generalization of PCA** !

It extends this idea using a Bayesian factor model for multiple omics.

PCA $\rightarrow Y \approx ZW^T$

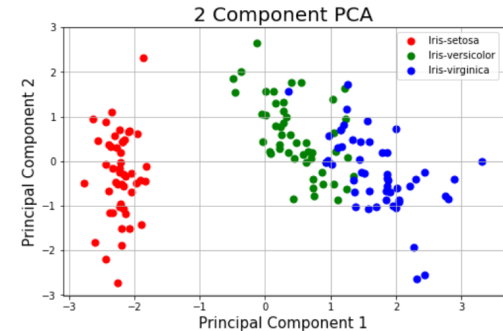
1 omic $\rightarrow$ **few latent components + weights**

MOFA $\rightarrow Y^{(1)} \approx ZW^{(1)T}$

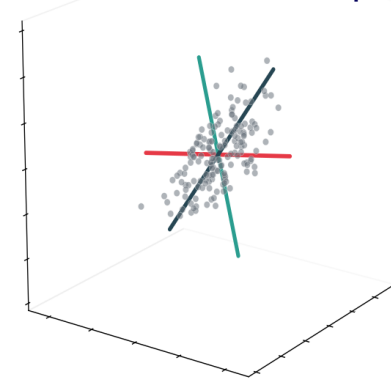
$Y^{(2)} \approx ZW^{(2)T}$

$Y^{(3)} \approx ZW^{(3)T}$

multiple omics $\rightarrow$ **few common latent components + omic-specific weights**



Multi-omics latent space



— Latent Factor 1 — Latent Factor 2 — Latent Factor 3



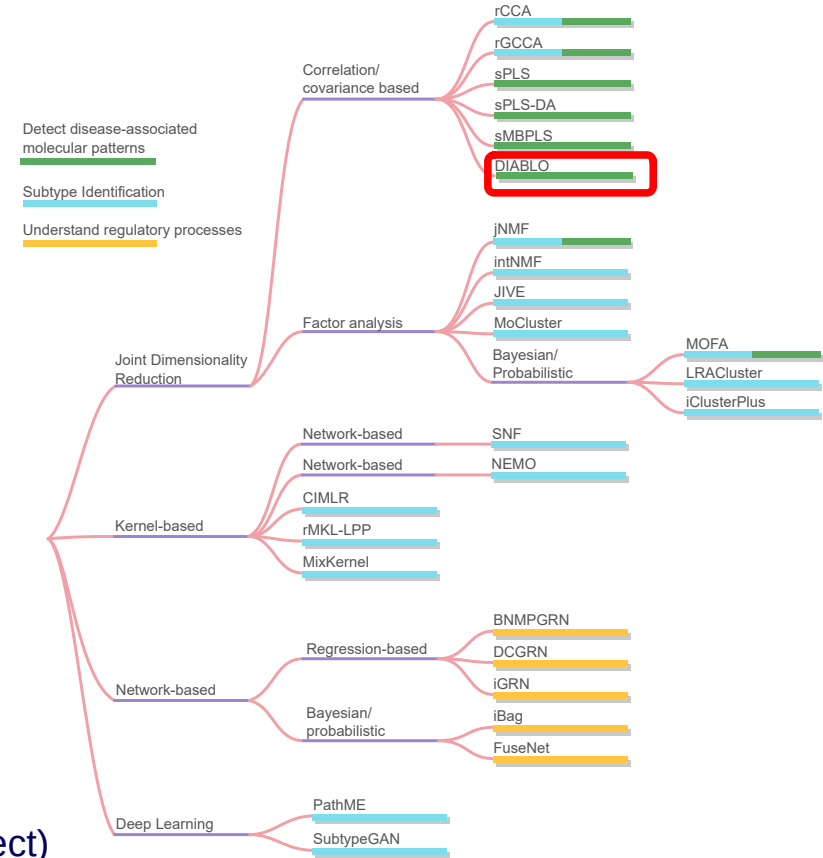
DIABLO (Singh A, et al. Bioinformatics, 2019)

DIABLO is a supervised extension of sGCCA (sparse Generalized Canonical Correlation Analysis), that identifies variables while discriminating phenotypic groups.

Objectives:

- 1) **maximize the covariance** between linear combinations of variables (latent components)
- 2) **maximize groups discrimination**

- Joint integration and supervised feature selection (LASSO-like)
- Identifies correlated multi-omics signatures associated with phenotype
- Sparse weights facilitate interpretation
- Low performance on non-linear data
- Training requires matched samples across omics
- Some hyperparameters need tuning (n components, n variable to select)





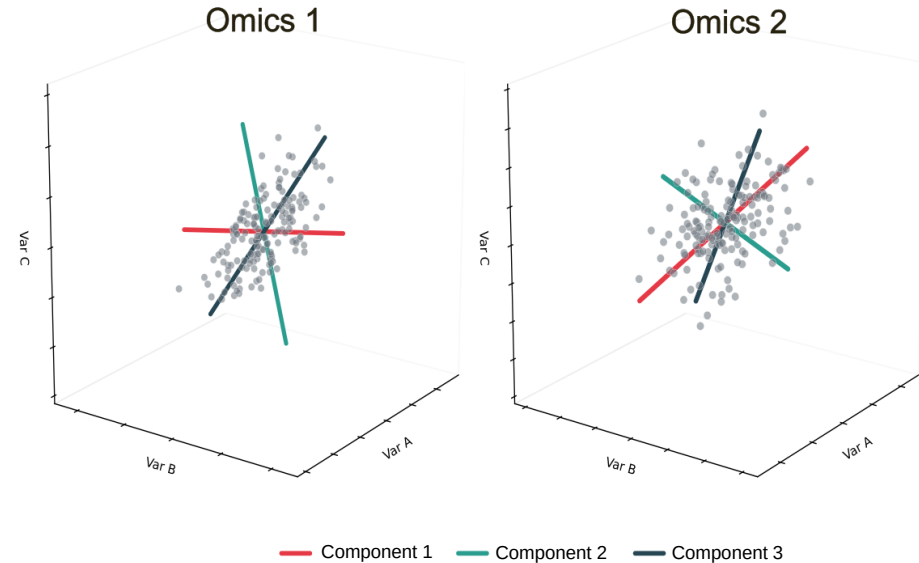
- For each omic, DIABLO learns a latent component as a **weighted combination of features**.

For sample i and omic q , we obtain the component score $t_i^{(q)}$ as:

$$t_i^{(q)} = a_1^{(q)}x_{i1}^{(q)} + a_2^{(q)}x_{i2}^{(q)} + \dots + a_p^{(q)}x_{ip}^{(q)}$$

- Sparsity is imposed on the weights**, so many coefficients can become zero (i.e. feature selection)
- Components are **learned jointly** to maximize association across connected omics while discriminating the phenotype.

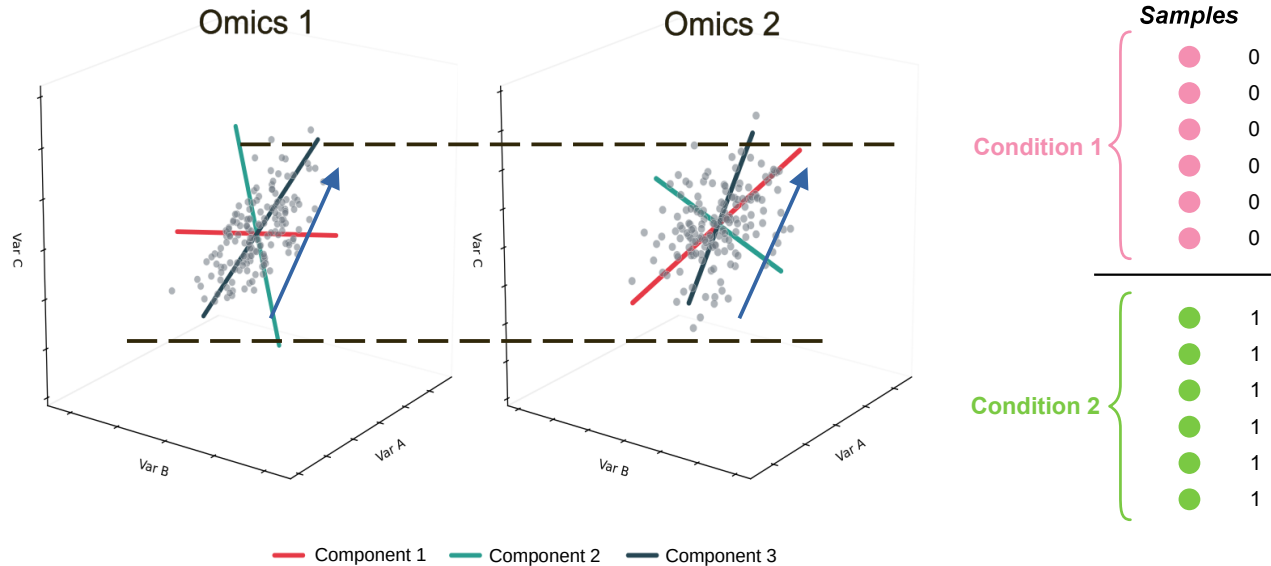
Each omic has its own components!





Dummy matrix → provides the supervised frame to select the components discriminating the two conditions

Design matrix → defines how strongly we expect pairs of omics to be related (high assoc. = 1, low assoc. = 0)



Dummy matrix

Samples	Cond. 1	Cond. 2
S ₁	1	0
S ₂	0	1
...	...	...
S _N	1	0

Design matrix

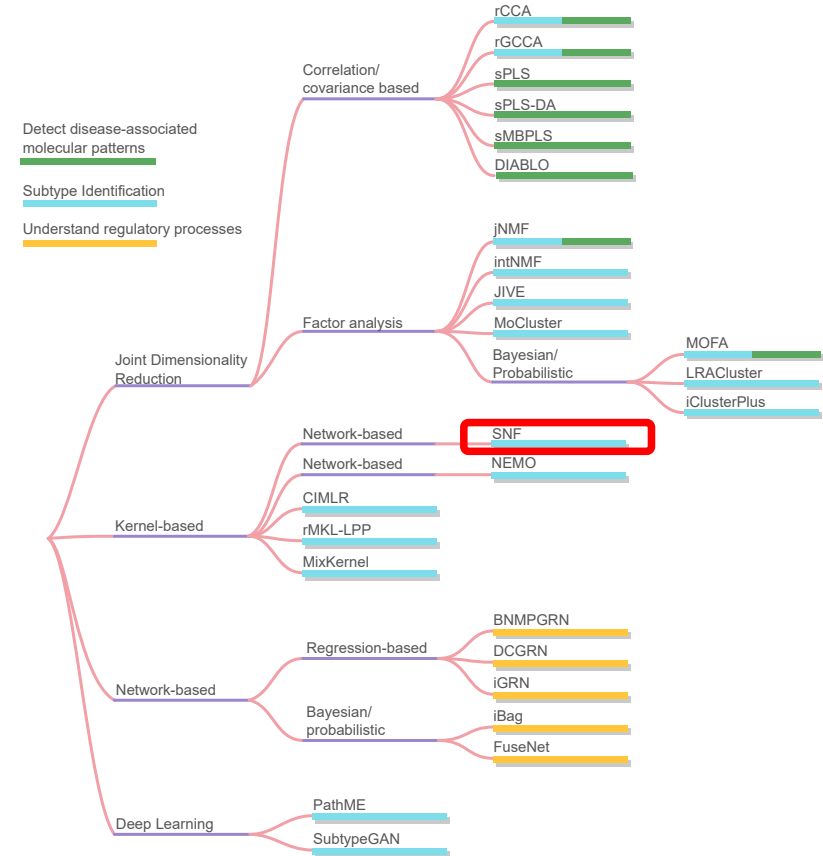
	RNA	Methylome
RNA	0	0.8
Methylome	0.8	0



Similarity Network Fusion (SNF) (Wang B, et al., Nature Methods, 2014)

SNF is an unsupervised multi-omics integration method that builds one sample similarity network per omic and iteratively fuses them into a single **integrated network** for **subtype discovery**, exploiting a message-passing process.

- Unsupervised subtype discovery
- High-dimensional features into an integrated sample-sample network
- Capture local/nonlinear relationships
- Network interpretation
- Limited biomarker interpretation
- Arbitrary choice of hyperparameters
- No missing-data handling
- Poor scaling with number of samples





How SNF works

1) **Omic-specific similarity matrix W** is computed.

2) Computation of **omic-specific global similarity matrix P** :

$$P(i, j) = \begin{cases} \frac{W(i, j)}{2 \sum_{k \neq i} W(i, k)} & , \text{ if } j \neq i \\ 1/2 & , \text{ if } j = i \end{cases}$$

3) Computation of the **omic-specific local similarity matrix S , retaining only k -nearest neighbours**:

$$S(i, j) = \begin{cases} \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)} & , \text{ if } j \in N_i \\ 0 & , \text{ otherwise} \end{cases}$$

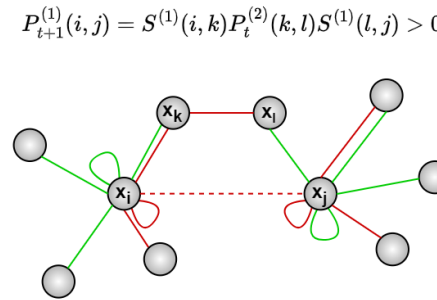
where $N_i = \{x_k | x_k \in kNN(x_i) \cup \{x_i\}\}$.

4) Integration through an **iterative parallel diffusion process and final average** (two views $s=\{1,2\}$):

$$\begin{aligned} P_{t+1}^{(1)} &= S^{(1)} \times P_t^{(2)} \times S^{(1)T} \\ P_{t+1}^{(2)} &= S^{(2)} \times P_t^{(1)} \times S^{(2)T} \end{aligned} \quad \longrightarrow \quad P_{t+1}^{(1)}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(1)}(i, k) P_t^{(2)}(k, l) S^{(1)}(l, j)$$

5) **Spectral clustering** for subgroup discovery.

Case I: indirect connection in $P^{(2)}$



$$P_{t+1}^{(1)}(i, j) = S^{(1)}(i, k) P_t^{(2)}(k, l) S^{(1)}(l, j) > 0$$

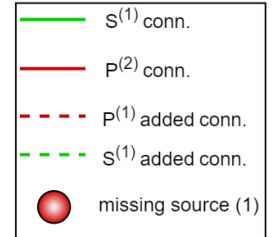


Image adapted from Gliozzo J, et al Bioinformatics, 2025



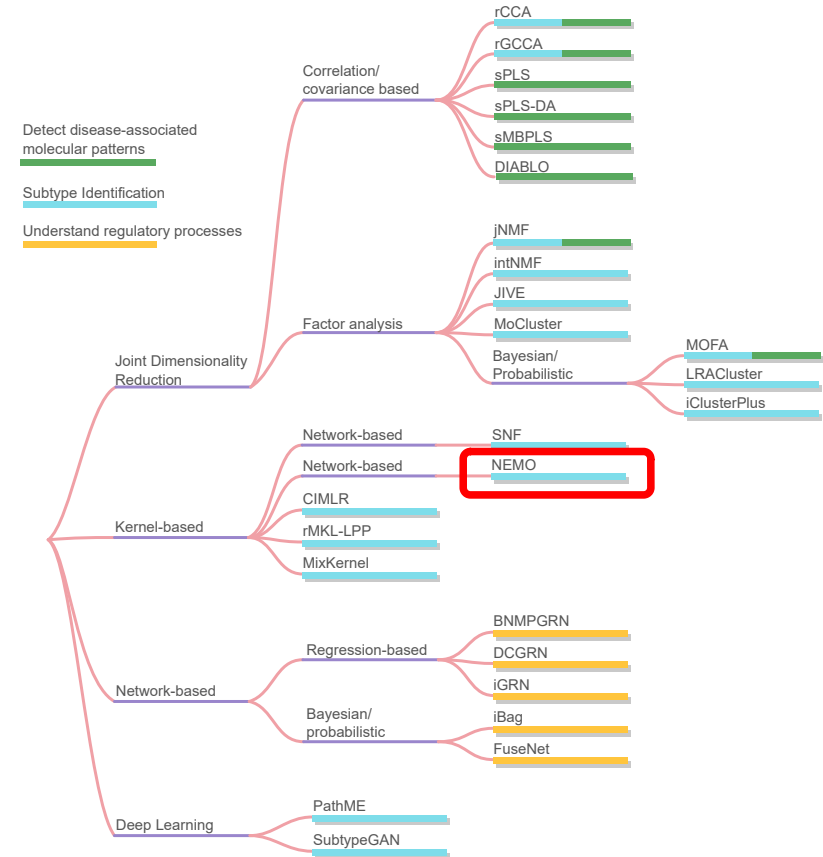
NEighborhood based Multi-Omics clustering (NEMO)

(Rappoport and Shamir, Bioinformatics, 2019)

NEMO is an **unsupervised network-based integration** approach that fuses local sample-neighbourhood similarities across omic-specific networks for **subtype discovery**.

- Unsupervised subtype discovery
- High-dimensional features into an integrated sample-sample network
- Capture local/nonlinear relationships
- Sample-sample network interpretation
- Handles partial multi-omics datasets

- Limited biomarker interpretation
- Choice of hyperparameters
- Assumes each pair of samples shares at least one measured omic
- Poor scaling with number of samples





How NEMO works

1) Omic-specific similarity matrix using “**scaled exponential Euclidean kernel**” (same as SNF).

2) Define a **relative similarity matrix** for each omic l :

$$RS_l(i, j) = \frac{S_l(i, j)}{\sum_{r \in \eta_{li}} S_l(i, r)} \cdot I(j \in \eta_{li}) + \frac{S_l(i, j)}{\sum_{r \in \eta_{lj}} S_l(r, j)} \cdot I(i \in \eta_{lj})$$

Similarity is evaluated relative to the local neighborhoods of both samples (i,j).

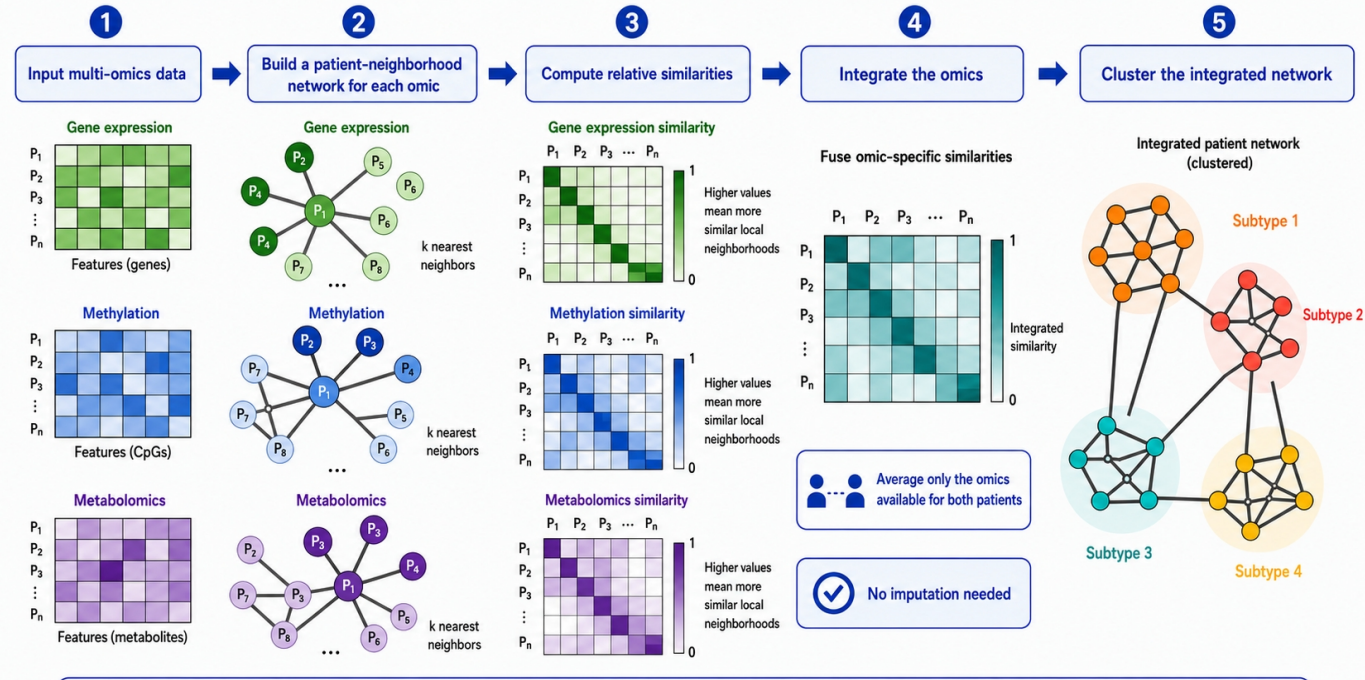
3) Integration by **average** considering **omics types available for both samples JM(i,j)**:

$$ARS(i, j) = \frac{1}{|JM(i, j)|} \sum_{l \in JM(i, j)} RS_l(i, j)$$

4) **Spectral clustering** for subgroup discovery.

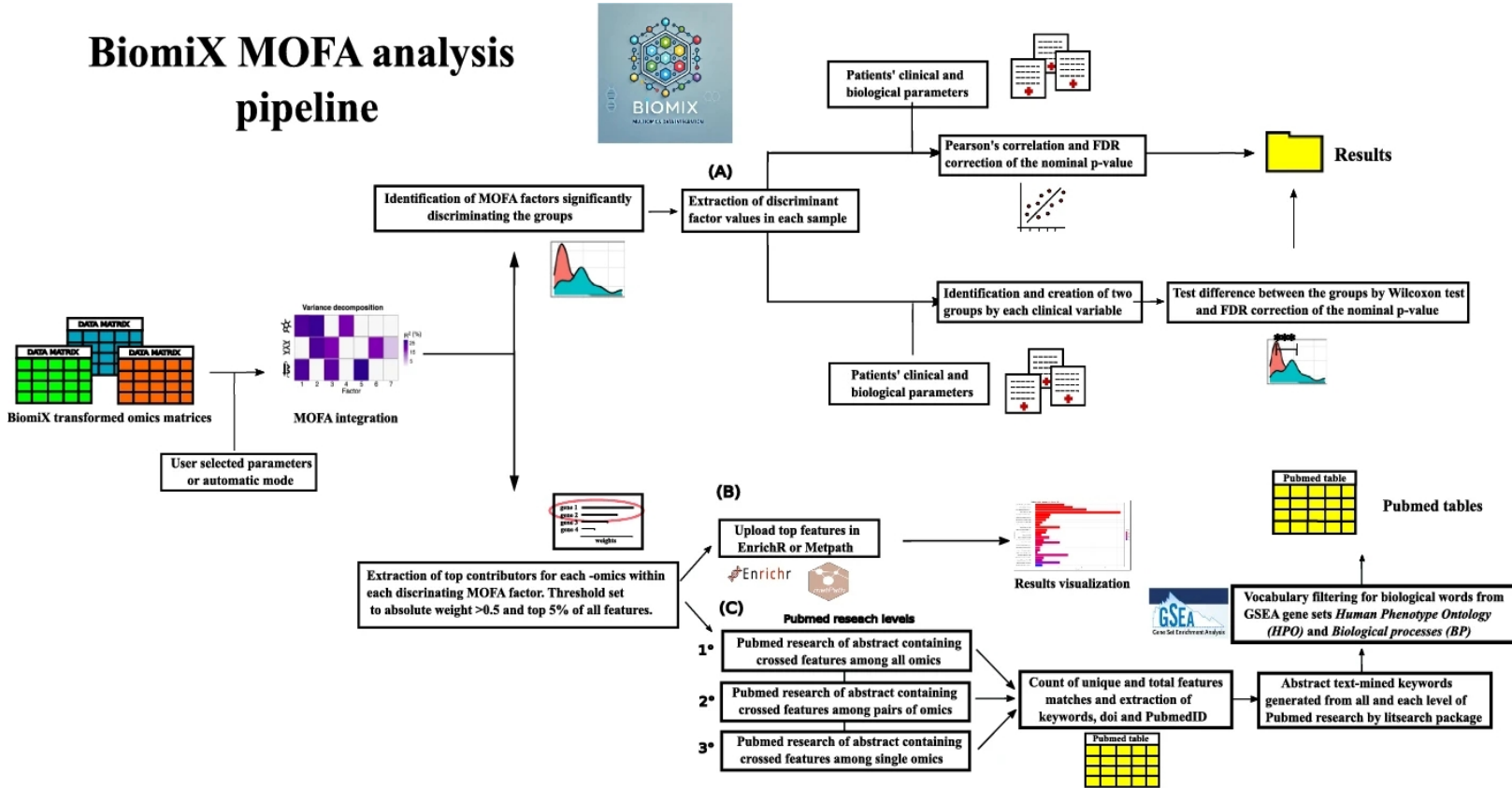
Unlike SNF: no iterative network diffusion.

Handles missing omics
Each pair of samples must share ≥ 1 measured omic





BiomiX MOFA analysis pipeline



What's new with BiomiX?

- Automatic selection MOFA/ DIABLO factors (stopping when new variance capture < 1%)
- Selection of factors separating the two conditions (semi-supervised)
- Automatic annotation of the latent factors
 - Correlation by clinical data
 - Pathway analysis
 - Bibliographic research



MOFA tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

Factor to explore

Omics overlay

Main interface

⚙ Setup

Advanced Options

📊 QC & Imputation

JSON Preview

General

Metabolomics

MOFA & Diabolo

SNF / NEMO

MintTea

Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Advance option MOFA



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diablo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

⚙ Setup **Advanced Options** 📊 QC & Imputation JSON Preview

General Metabolomics **MOFA & Diablo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Converging speed, the slower the more accurate

All features contribution to a factor are ranked from 0 to 1. Score threshold to filter features for interpretation analysis

Advance option MOFA



Biomix v2.5

Multiomics analysis pipeline



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

EGAS00001001746_metadata_CLL.xls

184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS

Condition (GROUP_1)

mutated

Control (GROUP_2)

unmutated

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

Select folder...

C:/Users/crist/Desktop/Biomix2.5

- 1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.
- 2) Set the mutated vs unmutated comparison
- 3) Output folder in Biomix 3.0 folder



OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Transcriptomics	☒	RNA	☐	Browse file... EGAS00001001746_transcriptomics.xls C:/Users/crist/Desktop/BiomiX2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls
Input 2	☐	☐	Methylomics	☒	Methy	☐	Browse file... EGAS00001001746_methylomics.xls C:/Users/crist/Desktop/BiomiX2.5/Methylomics/INPUT/EGAS00001001746_methylomics.xls
Input 3	☐	☐	Undefined	☐	Label input 3	☐	Browse file... No file selected

- 4) Load EGAS00001001746_transcriptomics.xls from Transcriptomics folder.
- 5) Load EGAS00001001746_methylomics.xls from Methylomics folder.
- 6) Click on integration to select the slot to integrate.



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

10

Factor to explore

1

Omics overlay

0

Select 10 factors

Advanced Options

▶ Run Biomix

Download JSON

✓ Ready to run





1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Each omics has the positive and negative contributors for the “discriminating” factors

Example:

	A	B	C
1	weights	features	abs_weight
2	-0.50266	cg07825294/DAZAP1	0.502661
3	-0.50448	cg07499142/PIK3CD	0.504481
4	-0.50682	cg01320698/PIK3CD	0.506823
5	-0.51647	cg03437605/RHOBTB	0.516468
6	-0.52646	cg11978120/NA	0.526455
7	-0.52949	cg14999716/HCFC2	0.529491
8	-0.53092	cg24545687/EP400	0.53092
9	-0.53151	cg09416521/FOXK2	0.53151

Factors_Associated_Articles	20/05/2026 08:48	Cartella di file	
Distribution_factors_contributions_mutat...	20/05/2026 08:56	Adobe Acrobat D...	24 KB
Methylomics_Methylomics_neg_factor_1.t...	20/05/2026 08:41	File TSV	13 KB
Methylomics_Methylomics_neg_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
Methylomics_Methylomics_pos_factor_1.t...	20/05/2026 08:41	File TSV	2 KB
Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
MOFA_MATRIX_mutated_vs_unmutated_1...	20/05/2026 08:40	File TSV	1 KB
MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB



1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Latent factor value (10 factors) per each sample. Dimensional reduction-like.

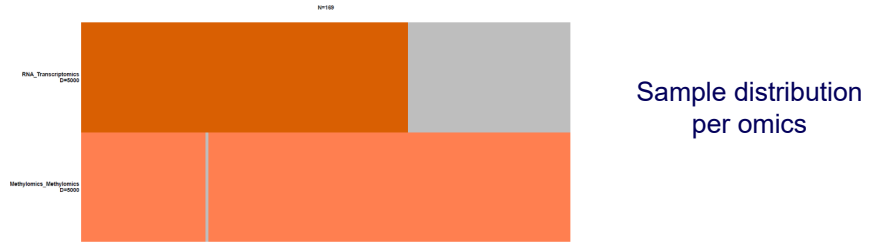
Variance explained by each factor within each omics

P.adjusted value of the “discrimination” power of each factor among the two conditions

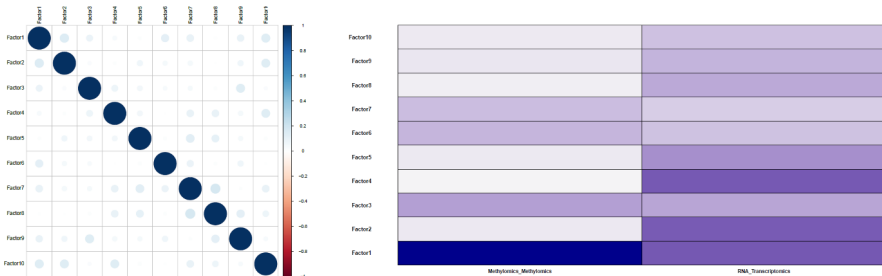
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Methylomics_Methylomics_neg_factor_1.t...	20/05/2026 08:41	File TSV	13 KB
Methylomics_Methylomics_neg_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
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Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
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MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB



1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors



Sample distribution
per omics

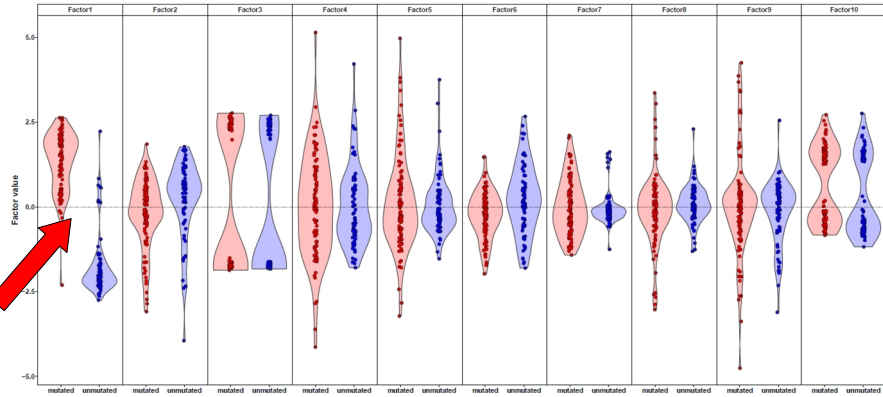


Factor co-
correlation test

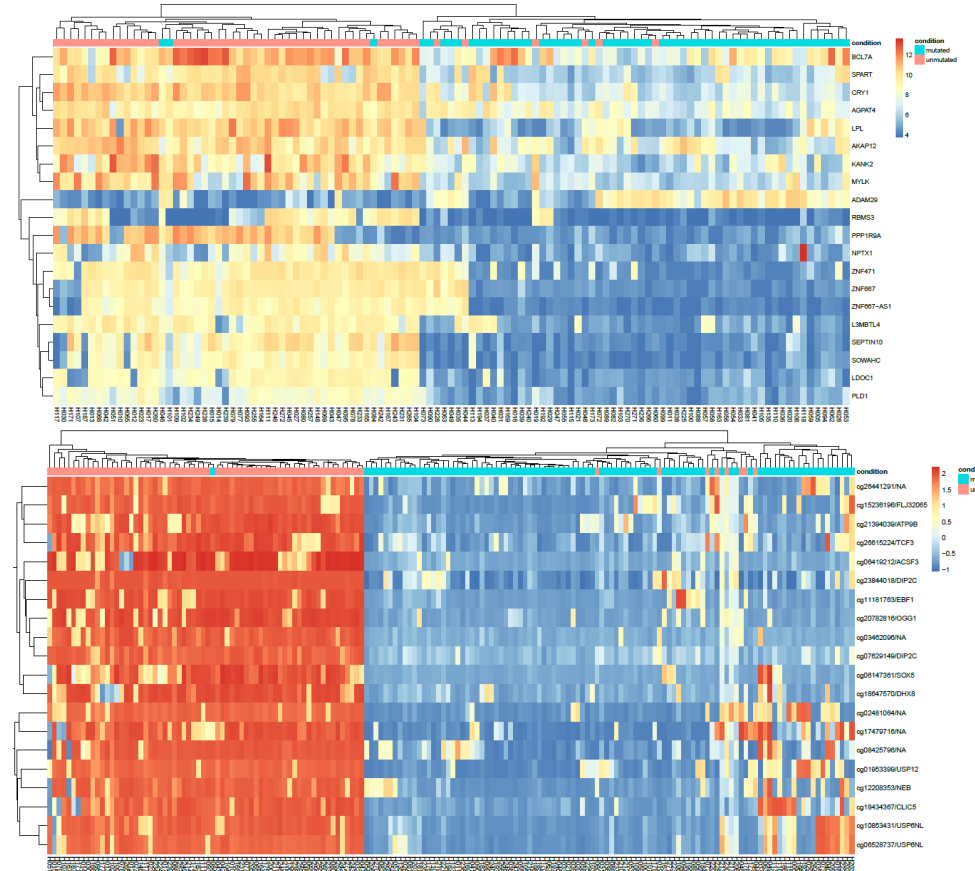
Variance explained per omics
and factor

Main pdf results including

Factors_Associated_Articles	20/05/2026 08:48	Cartella di file	
Distribution_factors_contributions_mutat...	20/05/2026 08:56	Adobe Acrobat D...	24 KB
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Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
Methylomics_Methylomics_pos_factor_1.t...	20/05/2026 08:41	File TSV	2 KB
Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
MOFA_MATRIX_mutated_vs_unmutated_1...	20/05/2026 08:40	File TSV	1 KB
MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB



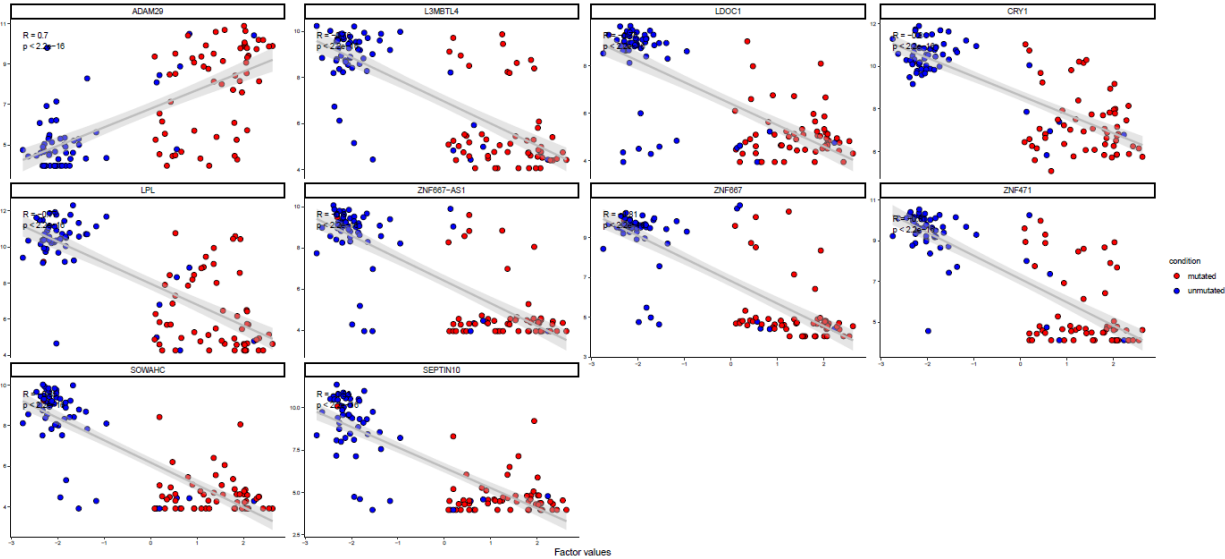
Violin plot factor separation per condition



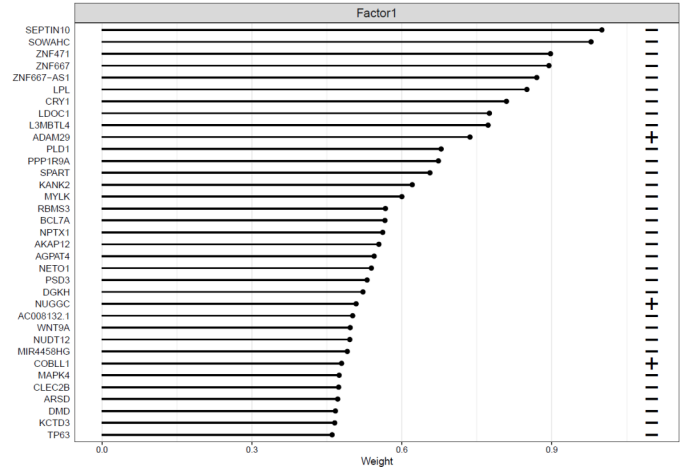
Heatmap to features
factor per omics



1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors



Top contributor discriminatory power per condition



Top contributor weights.



Nice, but what is behind
factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

🔍 FACTOR INTERPRETATION – USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type:

N° articles retrieved:

N° top Keywords extracted:

N° top contributors:

PATHWAY ANALYSIS

P.adj threshold:

N° pathways shown:

CLINICAL DATA INTEGRATION

Data types to correlate:

- ☒ Numerical metadata
- ☒ Binary metadata

Numerical clinical data file (optional): [EGAS00001001746_clinical_numeric.xlsx](#)

Binary clinical data file (optional): [EGAS00001001746_clinical_binary.xls](#)



Nice, but what is behind factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

FACTOR INTERPRETATION - USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type: Title/Abstract (dropdown)
N° articles retrieved: 10 (dropdown)
N° top Keywords extracted: 100 (dropdown)
N° top contributors: 10 (text input)

PATHWAY ANALYSIS

P.adj threshold: 0,05 (text input)
N° pathways shown: 10 (text input)

CLINICAL DATA INTEGRATION

Data types to correlate:
☒ Numerical metadata
☒ Binary metadata

Numerical clinical data file (optional):
Browse... [EGAS00001001746_clinical_numeric.xlsx](#)

Binary clinical data file (optional):
Browse... [EGAS00001001746_clinical_binary.xls](#)

1) Re-run including Numeric and binary clinical data. These will be correlated and tested with the MOFA factors calculated.

Are they associated?



Open:BiomiX2.5\Clinical_data\OUTPUT\Clinical_mutated_vs_unmutated_10_factors



	A	B	C	D	E
1	clinic	means	pval	p.adj	
2	IGHV.Uppsala.U.M.1	3.108624	5.3918365	4.7448162020209e-23	
3	del11q22.3	-1.86591	4.7366701	2.08413488138209e-05	
4	BRAF	-1.7882	0.001813	0.053191	
5	TP53	-1.12497	0.002874	0.063222	
6	LAMA1	-2.38988	0.007647	0.134582	
7	KLHL6	2.10113	0.016083	0.235885	
8	del13q14_bi	0.956899	0.019169	0.240977	
9	del9p21.3	-1.61911	0.036384	0.329951	
10	del17p13	-1.06866	0.035695	0.329951	
11	MED12	-1.67928	0.037494	0.329951	

1) Conclusion:
Factor 1 is directly
associated to the IGHV
mutation measurement



DIABLO tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



INTEGRATION OPTIONS

INTEGRATION
☒ Yes ☐ No

METHOD
☐ MOFA
☒ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS
N° factors (0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

Main interface DIABLO

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General Metabolomics **MOFA & Diabolo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

DIABLO PARAMETERS

Feature selection

N° iterations

Design matrix file (optional)
 Nessun file (opzionale)

Advance option DIABLO



🔍 FACTOR INTERPRETATION — USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type

Title/Abstract ▼

N° articles retrieved

10 ▼

N° top Keywords extracted

100 ▼

N° top contributors

10

PATHWAY ANALYSIS

P.adj threshold

0,05

N° pathways shown

10

CLINICAL DATA INTEGRATION

Data types to correlate

☒ Numerical metadata

☒ Binary metadata

Numerical clinical data file (optional)

Browse...

📄 EGAS00001001746_clinical_numeric.xlsx

Binary clinical data file (optional)

Browse...

📄 EGAS00001001746_clinical_binary.xls

**Don't forget the Clinical data
for the interpretation!**



1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

Example:

	A	B	C
1	weights	features	abs_weight
2	-0.50266	cg07825294/DAZAP1	0.502661
3	-0.50448	cg07499142/PIK3CD	0.504481
4	-0.50682	cg01320698/PIK3CD	0.506823
5	-0.51647	cg03437605/RHOBTB	0.516468
6	-0.52646	cg11978120/NA	0.526455
7	-0.52949	cg14999716/HCFC2	0.529491
8	-0.53092	cg24545687/EP400	0.53092
9	-0.53151	cg09416521/FOXK2	0.53151

Each omics has the positive and negative contributors for the “discriminating” factors

Clinical_correlation	25/08/2026 08:21	Cartella di file	
Factor_Pathways	25/08/2026 08:11	Cartella di file	
Factors_Associated_Articles	25/08/2026 08:13	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	25/08/2026 08:21	File TSV	39 KB
DIABLO_MATRIX_mutated_vs_unmutated...	25/08/2026 08:21	File TSV	1 KB
DIABLO_mutated_vs_unmutated.pdf	25/08/2026 08:21	Adobe Acrobat D...	677 KB
DIABLO_SEPARATION_mutated_vs_unmut...	25/08/2026 08:21	File TSV	2 KB
Distribution_factors_contributions_mutat...	25/08/2026 08:21	Adobe Acrobat D...	19 KB
RNA_Transcriptomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	8 KB
RNA_Transcriptomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB
RVA_Methylomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	11 KB
RVA_Methylomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB



1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

Latent factor value (10 factors) per each sample. Dimensional reduction-like.

Variance explained by each factor within each omics

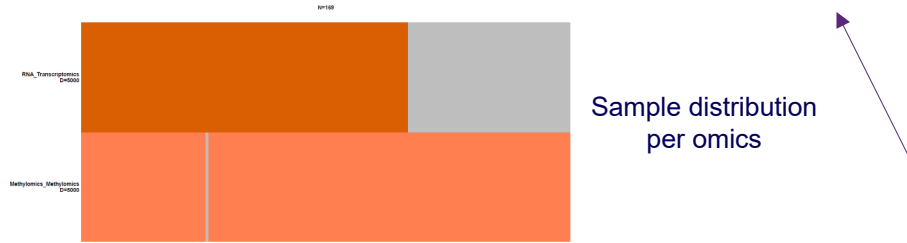
P.adjusted value of the “discrimination” power of each factor among the two conditions

Clinical_correlation	25/08/2026 08:21	Cartella di file	
Factor_Pathways	25/08/2026 08:11	Cartella di file	
Factors_Associated_Articles	25/08/2026 08:13	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	25/08/2026 08:21	File TSV	39 KB
DIABLO_MATRIX_mutated_vs_unmutated...	25/08/2026 08:21	File TSV	1 KB
DIABLO_mutated_vs_unmutated.pdf	25/08/2026 08:21	Adobe Acrobat D...	677 KB
DIABLO_SEPARATION_mutated_vs_unmut...	25/08/2026 08:21	File TSV	2 KB
Distribution_factors_contributions_mutat...	25/08/2026 08:21	Adobe Acrobat D...	19 KB
RNA_Transcriptomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	8 KB
RNA_Transcriptomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB
RVA_Methylomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	11 KB
RVA_Methylomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB

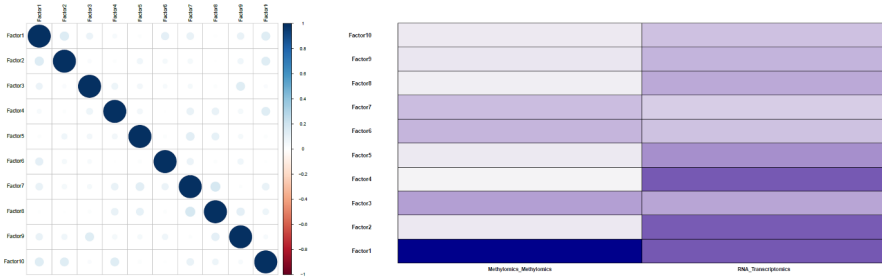


1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

Main pdf results including



Sample distribution
per omics



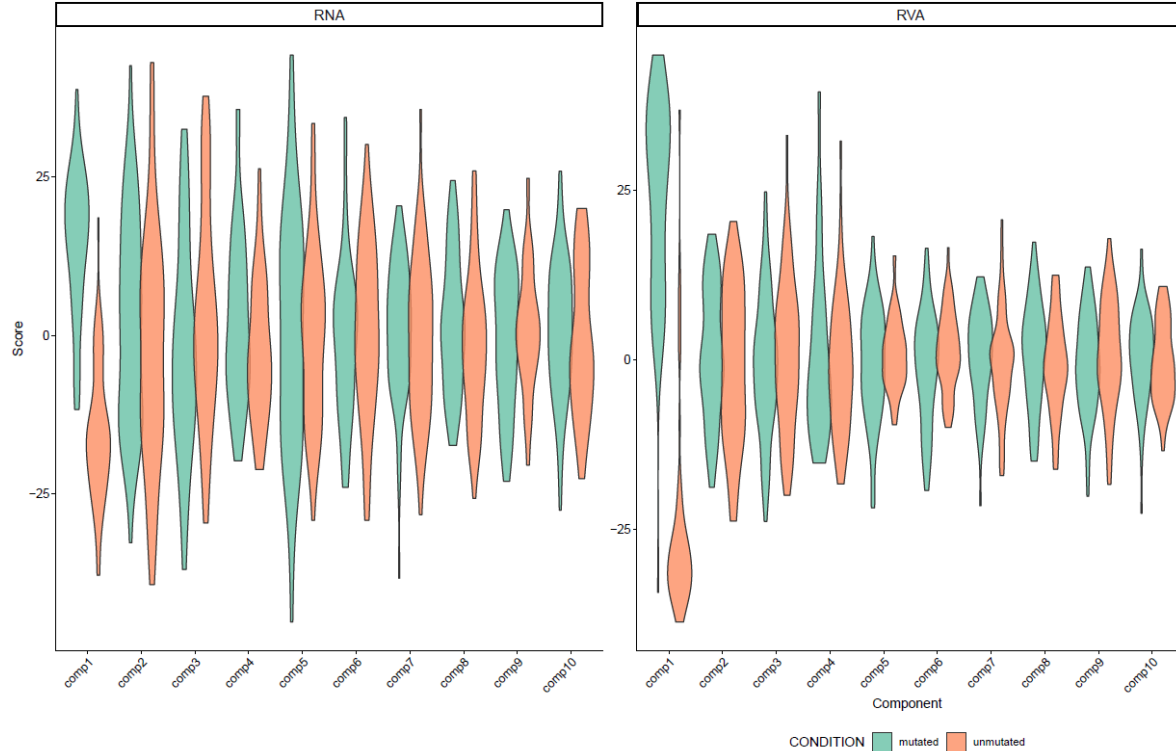
Factor co-
correlation test

Variance explained per omics
and factor

Clinical_correlation	25/08/2026 08:21	Cartella di file	
Factor_Pathways	25/08/2026 08:11	Cartella di file	
Factors_Associated_Articles	25/08/2026 08:13	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	25/08/2026 08:21	File TSV	39 KB
DIABLO_MATRIX_mutated_vs_unmutated...	25/08/2026 08:21	File TSV	1 KB
DIABLO_mutated_vs_unmutated.pdf	25/08/2026 08:21	Adobe Acrobat D...	677 KB
DIABLO_SEPARATION_mutated_vs_unmut...	25/08/2026 08:21	File TSV	2 KB
Distribution_factors_contributions_mutat...	25/08/2026 08:21	Adobe Acrobat D...	19 KB
RNA_Transcriptomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	8 KB
RNA_Transcriptomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB
RVA_Methylomics_neg_factor_1.tsv	25/08/2026 08:21	File TSV	11 KB
RVA_Methylomics_pos_factor_1.tsv	25/08/2026 08:21	File TSV	3 KB



DIABLO scores by component and condition

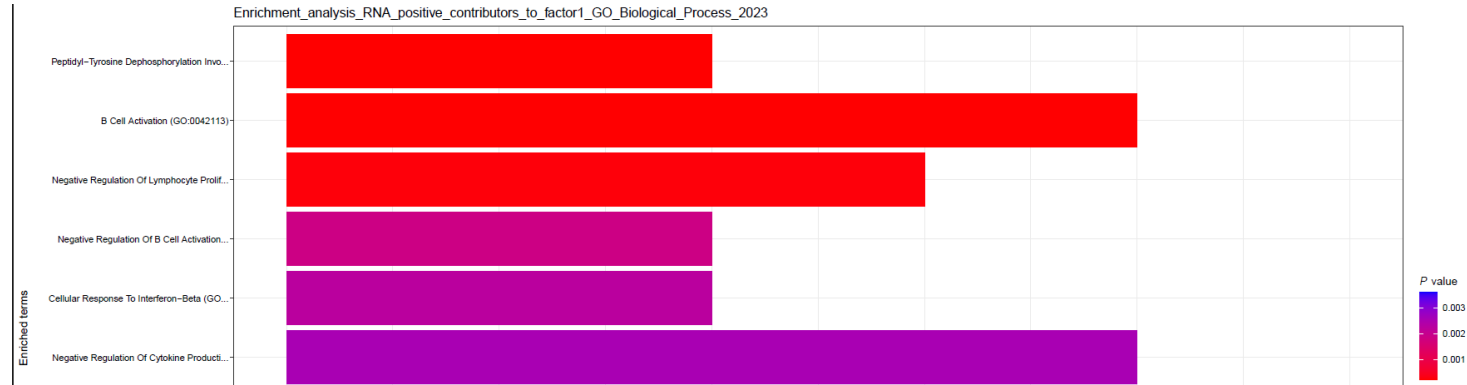


1) Factor 1 is discriminating patients by IGHV mutation measurement.

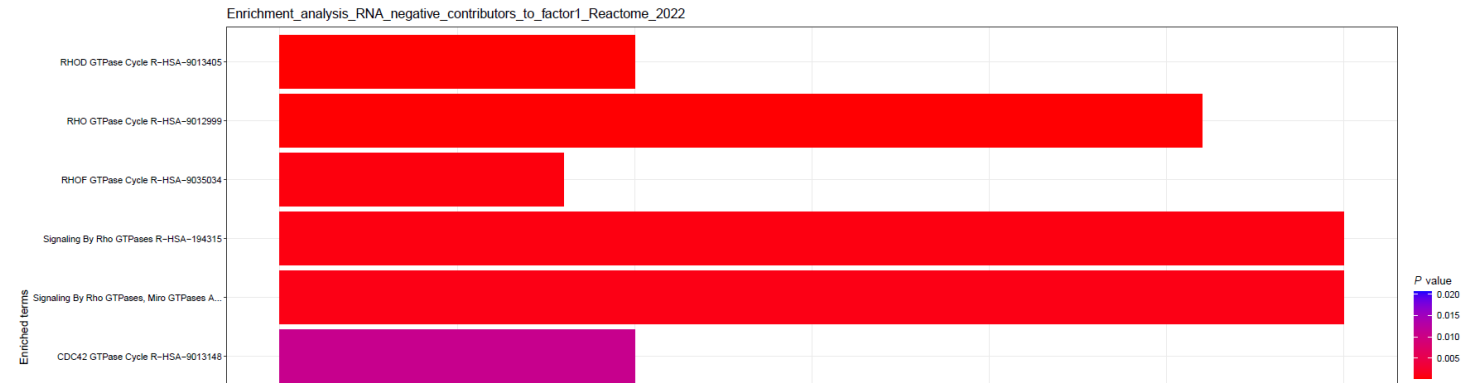
Which biological processes are behind it?



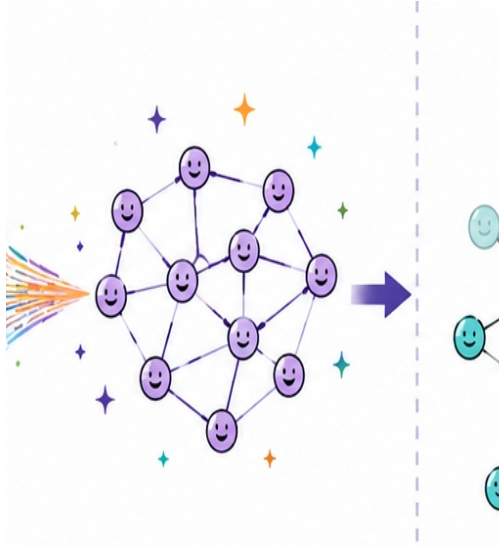
1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors\Factor_Pathways



1) Mutated patients are associated to higher values in factor 1 → Positive contributors



1) Unmutated patients are associated to Lower values in factor 1 → Negative contributors



SNF & NEMO Tutorials

Network-based multi-omics approaches



DATASET

Tutorial will use the **Chronic Lymphocytic Leukemia (CLL)** cohort (ID **EGAS00001001746**):

- *EGAS00001001746_metadata_CLL.tsv*:
metadata table having ID, DISEASE, AGE, GENDER, CONDITION, Treated, Alive
- *EGAS00001001746_methylomics.tsv*:
methylomic CpG sites (Illumina 450K array) profiles of the top-5000 most variant sites
- *EGAS00001001746_transcriptomics_VST.tsv*:
transcriptomic (RNA-seq) profiles of ≈ 54500 transcripts after VST (variance-stabilizing transformation)

CONDITION: IGHV-**mutated** (M-CLL) and IGHV-**unmutated** (U-CLL) chronic lymphocytic leukemia.

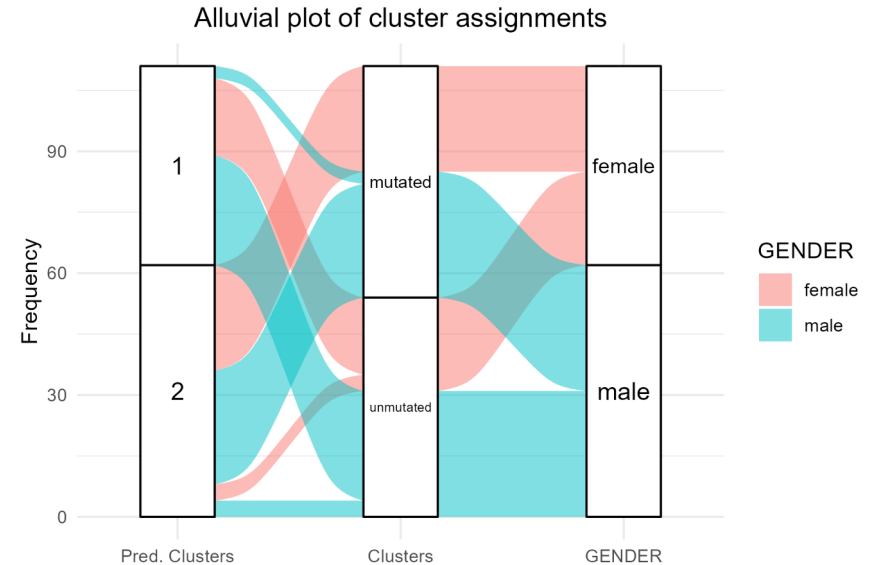


Number of clusters optimization SNF/NEMO

- Optimal number of clusters estimation: **eigengap/modified eigegap method**
- Internal clustering **validation metrics** (i.e. "Mean Silhouette Coefficient", "Dunn Index", "Entropy" and "Calinski and Harabasz Index")

Implemented interpretation methods SNF/NEMO

- Similarities heatmap for single omics and integrated matrix
- Network (graph) for single omics and integrated network
- External quality indices (i.e. ARI, AMI, NMI, NVI) if CONDITION (ground truth available)
- Post-hoc feature prioritization based on NMI
- Variables enrichment (chi-squared test for categorical, Kruskal-Wallis for continuous variables)
- Survival analysis (log-rank test and Kaplan-Meier plots)
- Alluvial plot comparing clustering, CONDITION (if any), variables of interest





1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

EGAS00001001746_metadata_CLL.tsv

184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS

Condition (GROUP_1)

mutated

Control (GROUP_2)

unmutated

3. OUTPUT FOLDER

Running in Docker: results will be saved to the folder you mounted on your machine via -v <your-folder>:/shared

/shared

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION (*RUM SINGLE OMICS BEFORE)	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Transcriptomics	☒	RNA	☐	Browse file... EGAS00001001746_transcriptomics_VST.tsv /shared/Example_dataset/EGAS00001001746/RNA_seq/EGAS00001001746_transcriptomics_VST.tsv
Input 2	☐	☒	Methylomics	☒	METHY	☐	Browse file... EGAS00001001746_methylomics.tsv /shared/Example_dataset/EGAS00001001746/Methylomics/EGAS00001001746_methylomics.tsv





INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☐ MOFA
☐ Diablo
☒ SNF
☐ NEMO
☐ MintTea

PARAMETERS

i Method-specific parameters for this integration method are in Advanced Options .

 Advanced Options

 Run Biomix

 Download JSON

 Ready to run



Advanced Options SNF

Model:

- Number of nearest neighbours
- Variance (controls decay of similarity with distance)
- Number of message-passing iterations

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo **SNF / NEMO** MintTea Metadata Filtering

SNF PARAMETERS

N° neighbors: 15 Variance affinity matrix: 0.5 N° iterations: 20

NEMO PARAMETERS

N° neighbors: 5 Only common samples: TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test: AGE GENDER Columns for Time (add .time) and survival column (add .event): Condition column: CONDITION

CLUSTERING AND FEATURE OPTIONS

Max cluster range: 6 N° variables in heatmap: 20 Max N° features SNF / NEMO: 5000 Scaling the SNF/NEMO data: TRUE



Let's explore the results!





Switch in the main interface to NEMO in the “Integration Options”
(we can avoid single-omics analysis since already run for SNF)

INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

☐ MOFA

☐ Diablo

☐ SNF

☒ NEMO

☐ MintTea

PARAMETERS

Method-specific parameters for this integration method are in [Advanced Options](#) .



Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo **SNF / NEMO** MintTea Metadata Filtering

SNF PARAMETERS

N° neighbors: 10 Variance affinity matrix: 0.5 N° iterations: 20

NEMO PARAMETERS

N° neighbors: 15 Only common samples: TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test: AGE GENDER Columns for Time (add .time) and survival column (add .event): Condition column: CONDITION

CLUSTERING AND FEATURE OPTIONS

Max cluster range: 6 N° variables in heatmap: 20 Max N° features SNF / NEMO: 5000 Scaling the SNF/NEMO data: TRUE

Advanced Options NEMO

Model:

- Number of nearest neighbours
- Consider only samples having all omics available

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)



Let's explore the results!





Issues & suggestions

BiomiX is an actively developed research project; if you encounter any problem or have suggestions, please open an issue on GitHub or contact the authors.

GitHub issues page link: <https://github.com/BiomiX-consortium/BiomiX2.5/issues>

Authors contacts:

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